

A top-down view of a desk with various items: a stethoscope, a laptop displaying a website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop, containing the course title and name. A faint watermark 'Licensed to Shreyas Kalunke' is visible diagonally across the image.

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SEXUALLY TRANSMITTED INFECTIONS (STIs)

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

CNS Central Nervous System

HIV Human Immunodeficiency Virus

HPV Human Papillomavirus

HSV Herpes Simplex Virus

LGV Lymphogranuloma venereum

MSM Men who have sex with men

NAAT Nucleic Acid Amplification Test

PID Pelvic Inflammatory Disease

STI Sexually Transmitted Infection

OBJECTIVES

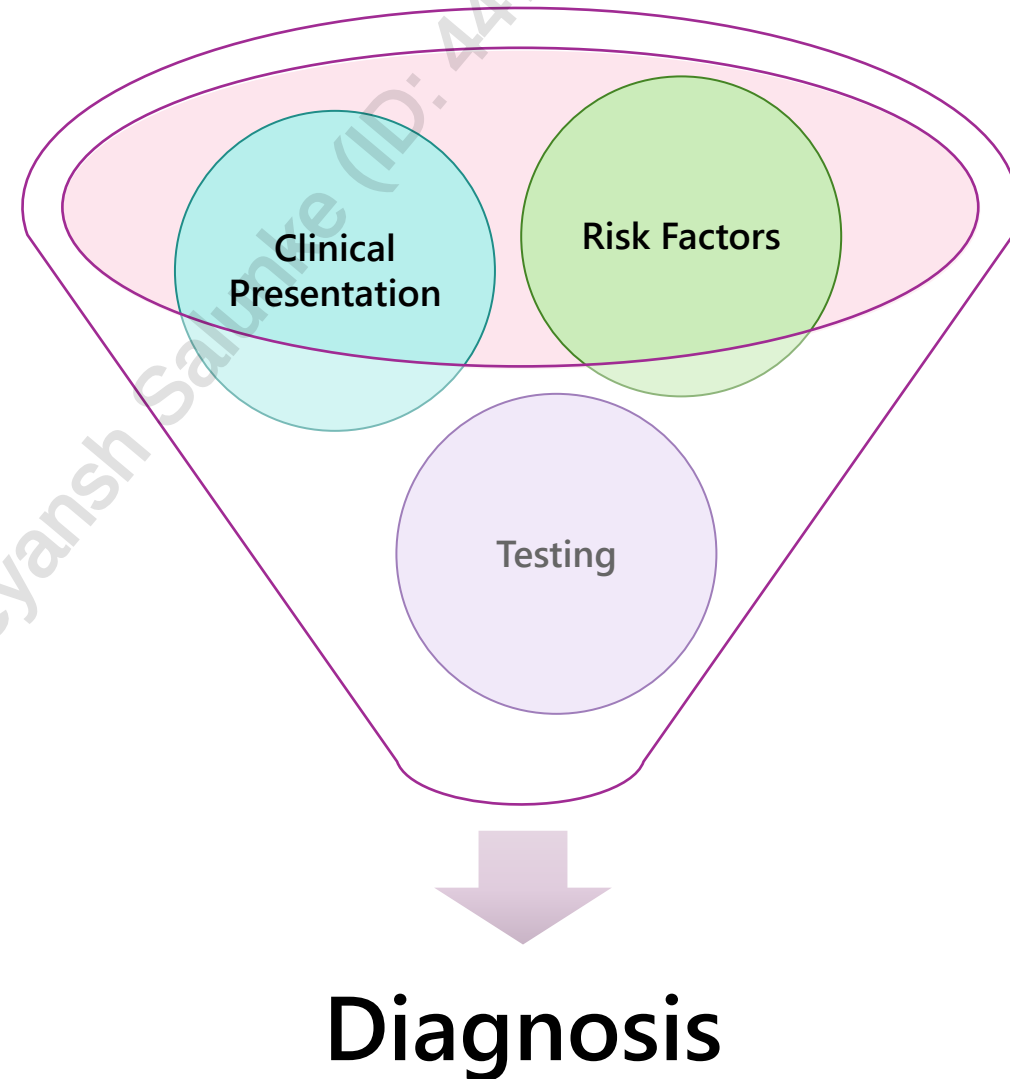
- Describe pathophysiology, risk factors, clinical presentation and organisms involved in the following STIs:
 - Gonorrhea
 - Chlamydia
 - Syphilis
 - Genital warts (HPV)
 - Chancroid
 - Pubic Lice and Scabies
 - Pelvic inflammatory disease (PID)
- Discuss and compare therapeutic options for managing all the STIs in terms of efficacy, adverse effects, and duration of therapy
- Provide appropriate counseling for STIs including preventative strategies



HIV, Hepatitis, HSV are covered in separate lectures

OVERVIEW

- Sexually transmitted infections share common risk factors and should be considered as potential co-infections
- Syndromic approach helpful to define likely pathogens and testing/treatment requirements



COMMON RISK FACTORS/RISK GROUPS

Number of sexual partners

Homelessness

Age < 25

Men who have Sex with Men (MSM)

Diagnosis with prior STI

Unprotected/unsafe sexual practices

Sex workers/ transactional sex

Illicit substance use

CLINICAL PRESENTATION

Cervicitis	Gonorrhea, chlamydia, trichomonas, herpes simplex virus (HSV)
Epididymitis	Chlamydia, gonorrhea, gram-negative bacteria
Genital ulcer disease	HSV, chlamydia (LGV), syphilis, H. ducreyi, K. granulomatis
Pelvic inflammatory disease	Chlamydia, gonorrhea, gram-negative bacteria, anaerobes
Prostatitis	Chlamydia, gonorrhea, gram-negative bacteria
Enteric infection	HSV, chlamydia, syphilis, Entamoeba histolytica, Salmonella spp., Shigella spp., Giardia lamblia
Urethritis	Chlamydia, gonorrhea, HSV, Trichomonas vaginalis, Mycoplasma/Ureaplasma spp.
Vaginal discharge	Bacterial vaginosis, Trichomonas vaginalis, Candida spp.

GONORRHEA

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CASE #1

SM is a 24-year-old male who presents to a walk-in clinic with a 3-day history of dysuria and purulent urethral discharge. He reports unprotected intercourse with a new partner one week ago. Nucleic acid amplification testing (NAAT) from a first-catch urine sample is positive for *Neisseria gonorrhoeae* and negative for *Chlamydia trachomatis*. He has no known drug allergies.

OVERVIEW

- One of the most urgent global antimicrobial resistance threats
- The causative organism is *Neisseria gonorrhoeae*
- The diagnosis involves culture testing and/or Nucleic Acid Amplification Technology (NAAT) of the affected clinical site(s)
- Clinical presentation:
 - Symptomatic or asymptomatic
 - Males initially present with purulent discharge associated with dysuria; discharge may become more profuse and blood-tinged as infection progresses
 - Females present asymptotically, if vaginal discharge a result of cervicitis; have a risk of developing PID; tend to present with co-existing infections
 - Can involve anogenital, pharyngeal and/or disseminated infection (note: treated differently)

PHARMACOLOGICAL MEASURES (NEW)

Preferred treatments (all populations)	• Ceftriaxone 500mg IM x 1*
Alternative treatments (pharyngeal)	• Cefixime + azithromycin
Alternative treatment (anogenital)	• Cefixime + doxycycline PO x 7 days
Alternative treatment	• Azithromycin PO + gentamicin IV/IM - OR • Gentamicin IV/IM x 1 dose + doxycycline PO x 7 days
If resistant to 3 rd gen cephalosporins	• Ertapenem

*Note: if unable to rule out chlamydia, also treat empirically for chlamydia

Duration of treatment is 1 dose unless otherwise specified

PHARMACOLOGICAL MEASURES- DISSEMINATED DISEASE TREATMENT

Endocarditis	• Ceftriaxone IV daily x 28 days + azithromycin PO x 1 dose
Meningitis	• Ceftriaxone IV daily x 10-14 days + azithromycin PO x 1 dose
Ophthalmia	• Ceftriaxone IV once + azithromycin PO x 1 dose
Arthritis	• Ceftriaxone IV daily x 7 days + azithromycin PO x 1 dose

MONITORING PARAMETERS

- **“Test of Cure”** is recommended after completion of therapy, especially in the following settings:
 - Pharyngeal infection
 - Cases not treated with ceftriaxone
 - Documented resistance
 - Re-exposure or treatment failure
 - Pregnancy
 - Infection in a child
 - Disseminated infection
- **Culture-based tests** should be done **at least 3 days after treatment**
- **NAAT tests** should be completed **3-4 weeks after treatment**
(to prevent false-positive results due to the presence of non-viable organisms)
- **Repeat screening for individuals with a gonococcal infection is recommended 6 months after treatment**

THERAPEUTIC TIPS

- Sexual partners within 60 days need to be treated even if asymptomatic
 - If not sexually active within 60 days, the most recent sexual partner should be treated, especially if the partner is pregnant
- Abstain for sexual intercourse for 7 days after treatment is started
- Use latex or urethane condoms for prevention
- Spermicidal nonoxynol-9, topical antibacterial agents, douching, washing and urinating after intercourse do not prevent STI transmission
- Screen for sexual health history and other STIs (note, treatment for chlamydia included in gonorrhea regimen)

CASE #1

SM is a 24-year-old male who presents to a walk-in clinic with a 3-day history of dysuria and purulent urethral discharge. He reports unprotected intercourse with a new partner one week ago. Nucleic acid amplification testing (NAAT) from a first-catch urine sample is positive for *Neisseria gonorrhoeae* and negative for *Chlamydia trachomatis*. He has no known drug allergies.

What treatment would you advise for this patient? How would you suggest partner notification and treatment be carried out?

CHLAMYDIA

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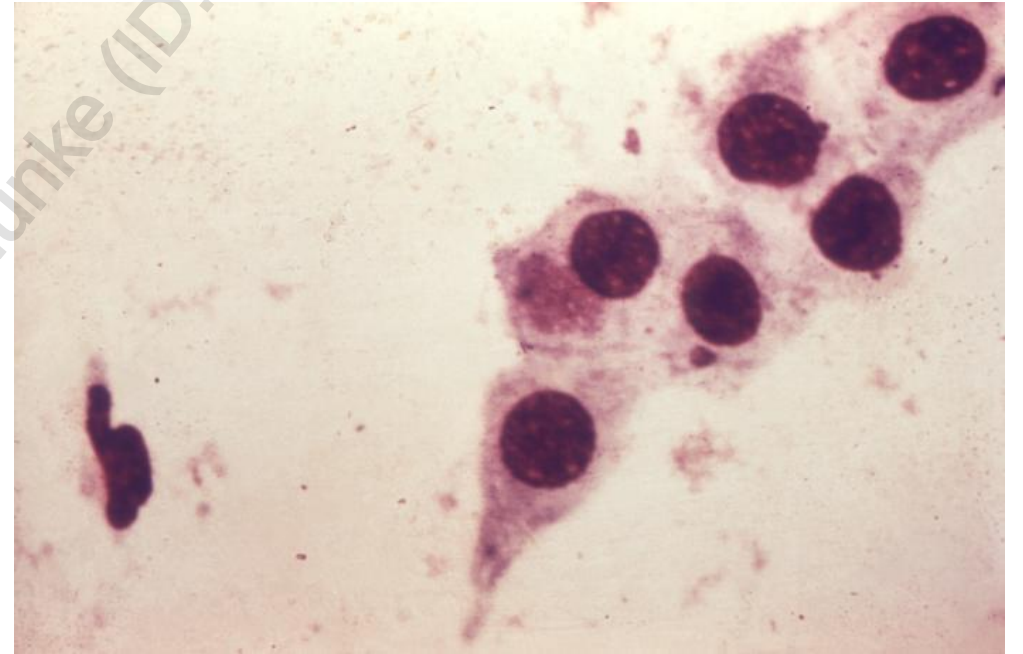
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CASE #2

SJ, a 21-year-old female, presents to a sexual health clinic for routine screening. She reports no symptoms but has had multiple new sexual partners in the past year. A NAAT from a vaginal swab returns positive for *Chlamydia trachomatis*. She has no allergies and is not pregnant.

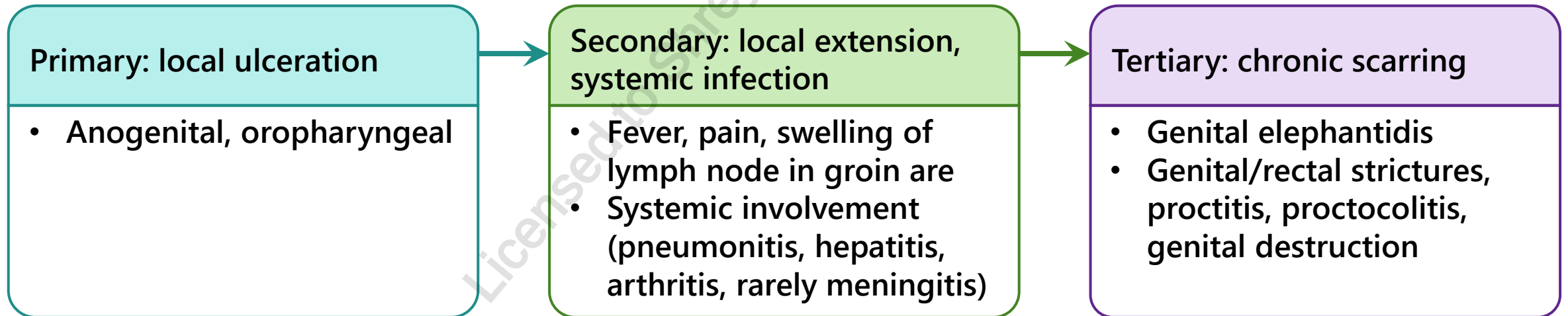
OVERVIEW

- Caused by *Chlamydia trachomatis*, obligate intracellular parasite
- Chlamydial genital infections are one the most common and underdiagnosed infectious diseases
- Asymptomatic to severe disease including Lymphogranuloma venereum (LGV, associated with several serovars of *C. trachomatis*) produces potentially invasive infections
- Testing via **NAAT**, less likely culture-based



CLINICAL PRESENTATION

- Uncomplicated chlamydial infections often asymptomatic
- Incubation period: 2-3 weeks (more variable for LGV)
- Females: cervicitis, vaginal discharge, dysuria, lower abdominal pain, abnormal vaginal bleeding, dyspareunia, conjunctivitis
 - Urethritis, Bartholinitis, endometritis, salpingitis, perihepatitis
- Males: urethral discharge, urethritis, dysuria, testicular pain, conjunctivitis
 - Epididymitis, prostatitis, proctitis, Reiter's syndrome
- Lymphogranuloma Venereum (LGV): 3 stages



PHARMACOLOGICAL MEASURES (NEW)

Preferred therapy (uncomplicated chlamydia)	• Azithromycin PO x 1 dose OR • Doxycycline BID x 7 days
Preferred therapy (pregnancy)	• Amoxicillin PO TID x 7 days OR • Azithromycin PO x 1 dose
LGV: preferred therapy	• Doxycycline PO x 21 days
LGV: alternative therapy	• Azithromycin PO weekly x 3 weeks

MONITORING PARAMETERS

- “Test of Cure” is NOT recommended after completion of therapy, unless
 - Adherence concerns
 - Alternative therapy used
 - Concerns for child abuse
 - Pregnancy
- NAAT tests should be completed 3-4 weeks after treatment
(to prevent false-positive results due to the presence of non-viable organisms)
- Repeat screening for individuals with chlamydia infection is recommended 3 months after treatment

THERAPEUTIC TIPS

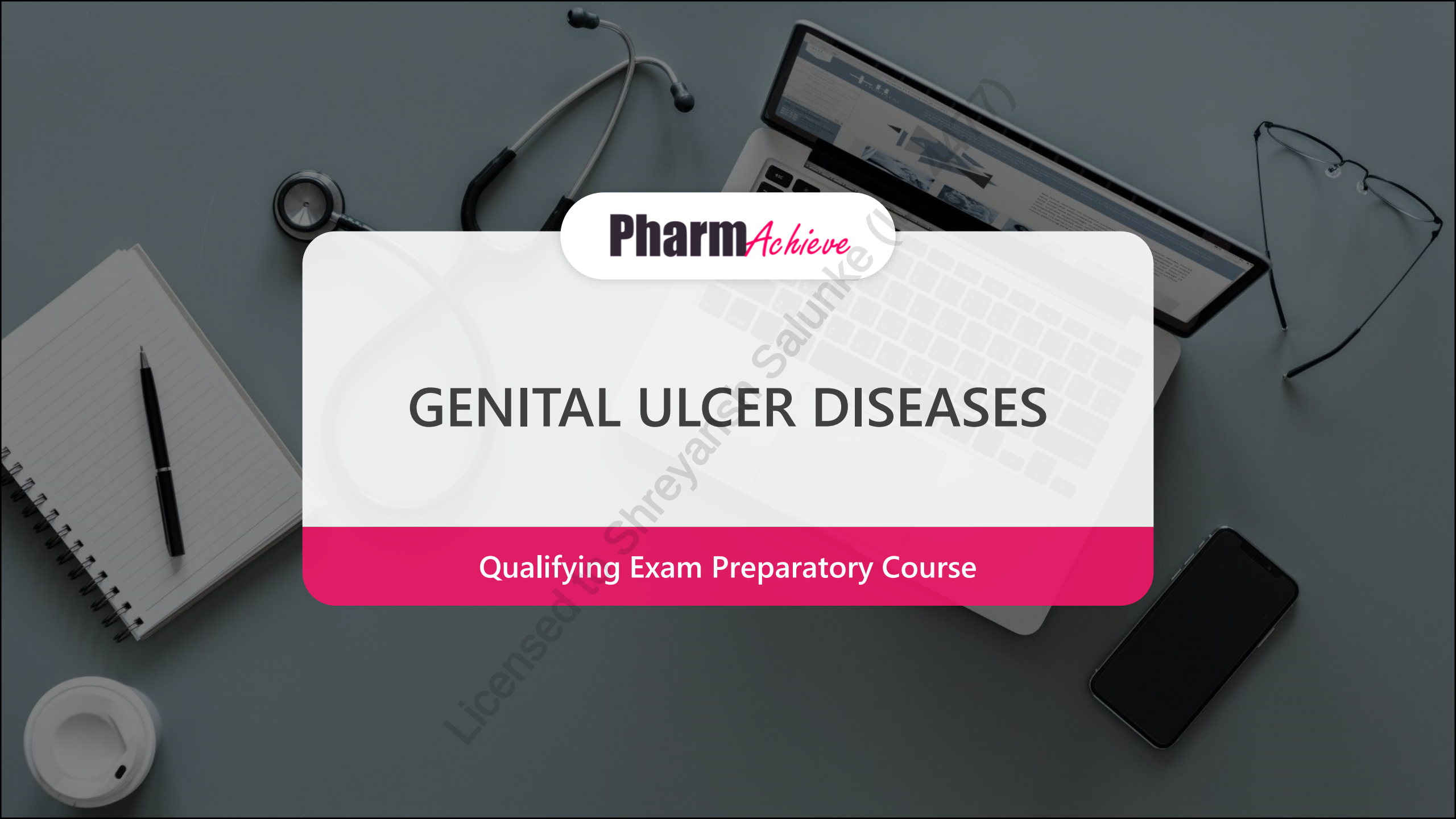
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- Screen for sexual health history and other STIs (note, treatment for chlamydia included in gonorrhea regimen)

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What treatment advice would you offer?

What are the suggested monitoring parameters?

A top-down view of a dark grey desk with various items: a stethoscope, a laptop displaying a medical website, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup lid. A large white rounded rectangle is centered on the desk, containing the course title and logo.

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GENITAL ULCER DISEASES

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SYPHILIS

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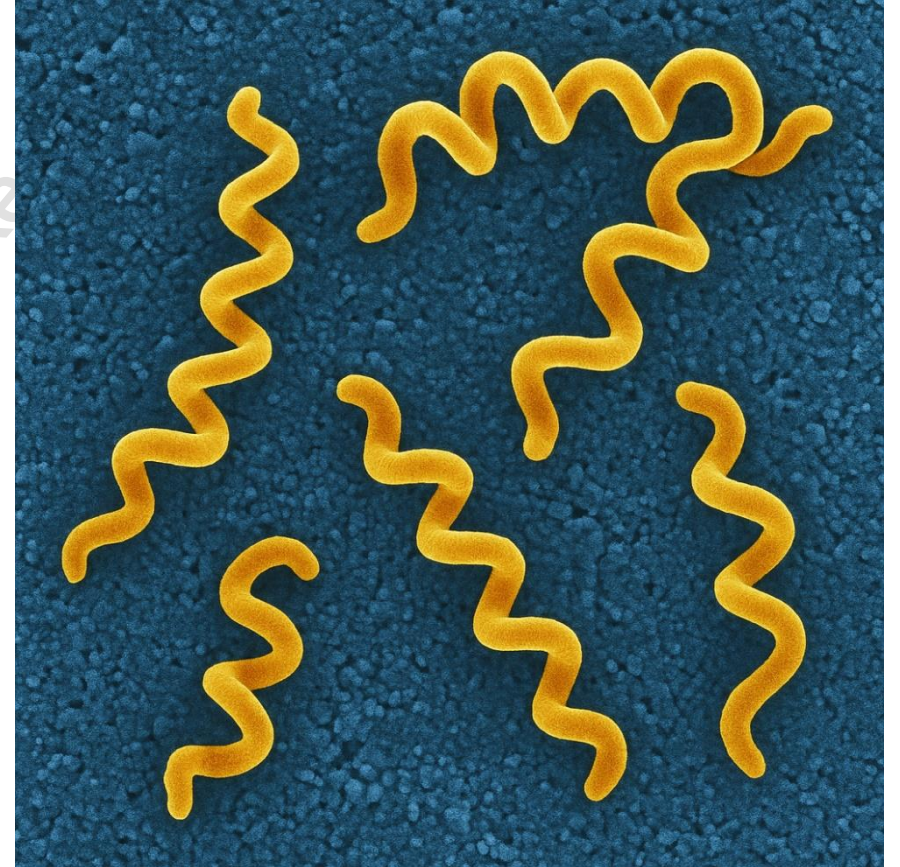
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CASE #3

KR, a 36-year-old male who has sex with men, presents with a painless genital ulcer that appeared one week ago. He has no systemic symptoms. Physical exam confirms a solitary, non-tender chancre on the shaft of the penis. Serologic testing with both treponemal and non-treponemal tests (e.g., EIA and RPR) supports a diagnosis of primary syphilis.

OVERVIEW

- Syphilis is caused by the spirochete, *Treponema pallidum*
- *Treponema pallidum* is acquired by:
 - Sexual contact with infected mucous membranes or cutaneous lesions
 - Accidental inoculation
 - Blood transfusion
- Rates of infection increasing (MSM>M:F)



CLINICAL PRESENTATION

Primary

- The primary stage, is characterized by the appearance of a painless chancre on cutaneous or mucocutaneous tissue at the original site of infection
- Initially a firm, round and painless papule that becomes ulcerated and indurated
 - Ulcer is filled with spirochetes
 - Chancre usually involves: penis in heterosexual males; penis or anus in homosexual males; vulva, perineum, or cervix of the female; lip or tongue can also be involved

Secondary: (2 weeks-6 months)

- Rash, fever, malaise, swollen lymph nodes
- Untreated patients will have skin lesions, widespread distribution
- Signs and symptoms may disappear within weeks or repeatedly reoccur and resolve for up to a year
- Macular syphilis is earliest manifestation of secondary syphilis

Tertiary

- If untreated, complications involving the skin, bones, CNS, heart, blood vessels, liver, skin/soft tissue and ophthalmic symptoms may appear years after infection

PHARMACOLOGICAL MEASURES

Preferred therapy (early stage: primary, secondary, or latent under 1 year)	• If true penicillin allergy AND failed alternative treatments: patient must be desensitized • Benzathine penicillin G IM x 1 dose
Alternative therapy (early stage: primary, secondary, or latent under 1 year)	• Doxycycline PO x 14 days OR • Ceftriaxone IM/IV x 10 days
Preferred therapy (late* stage: over 1-year, unknown duration, late-latent)	• Benzathine penicillin G IM x 3 doses QWEEKLY
Alternative therapy (late*-stage: over 1-year, unknown duration, late-latent)	• Doxycycline PO x 28 days OR • Ceftriaxone IM/IV x 10 days OR • If true penicillin allergy + failed alternative treatments: patient must be desensitized

*CSF analysis required to rule out neurosyphilis

PHARMACOLOGICAL MEASURES

Preferred therapy (CNS/ocular syphilis)	• Penicillin preferred, consider desensitization if allergic • Penicillin G IV x 10-14 days
Alternative therapy (CNS/ocular syphilis)	• Ceftriaxone IV x 10-14 days OR • Benzathine penicillin IM + probenecid PO x 10-14 days

Syphilis in pregnancy:

- Treated according to stage (early latent may be treated with 2 doses of benzathine penicillin weekly)
- Risk of transmission in untreated pregnant women is 70-100% with primary or secondary stages, 40% with early latent and 10% in late latent stages
- 40% of pregnancies with infectious syphilis results in fetal death

Jarisch-Herxheimer reaction:

- In patients with any stage of disease but common in tertiary
- Uterine contractions, acute febrile illness, myalgia, chills, skin rash and a reduced fetal heart rate that usually resolves 24 hours after treatment
- Can be treated with antipyretics or in severe cases corticosteroids and sedation

MONITORING PARAMETERS

- “Test of Cure” (serology) is recommended after completion of therapy variable based on stage of disease and HIV co-infection but generally 3, 6, 12 +/- 24 months. In pregnancy, start 1 month following treatment.
- Follow up for other conditions (pregnancy) or if symptoms return/persist

THERAPEUTIC TIPS

- Sexual partners require assessment and treatment relative to the stage of the incident case
 - Primary syphilis: partners within 3 months
 - Secondary syphilis: partners within 6 months
 - Early latent: partners within 1 year
 - Late latent: variable
- **Abstain from sexual intercourse for 7 days after completing therapy**
- Use latex or urethane condoms for prevention

CASE #3

KR, a 36-year-old male who has sex with men, presents with a painless genital ulcer that appeared one week ago. He has no systemic symptoms. Physical exam confirms a solitary, non-tender chancre on the shaft of the penis. Serologic testing with both treponemal and non-treponemal tests (e.g., EIA and RPR) supports a diagnosis of primary syphilis. **He has a self-reported beta-lactam allergy, which is described as nausea/vomiting in childhood.**

What is the recommended therapy for this patient?

What would you monitor?

What if he had neurosyphilis?

HUMAN PAPILLOMAVIRUS (HPV)/ ANOGENITAL WARTS

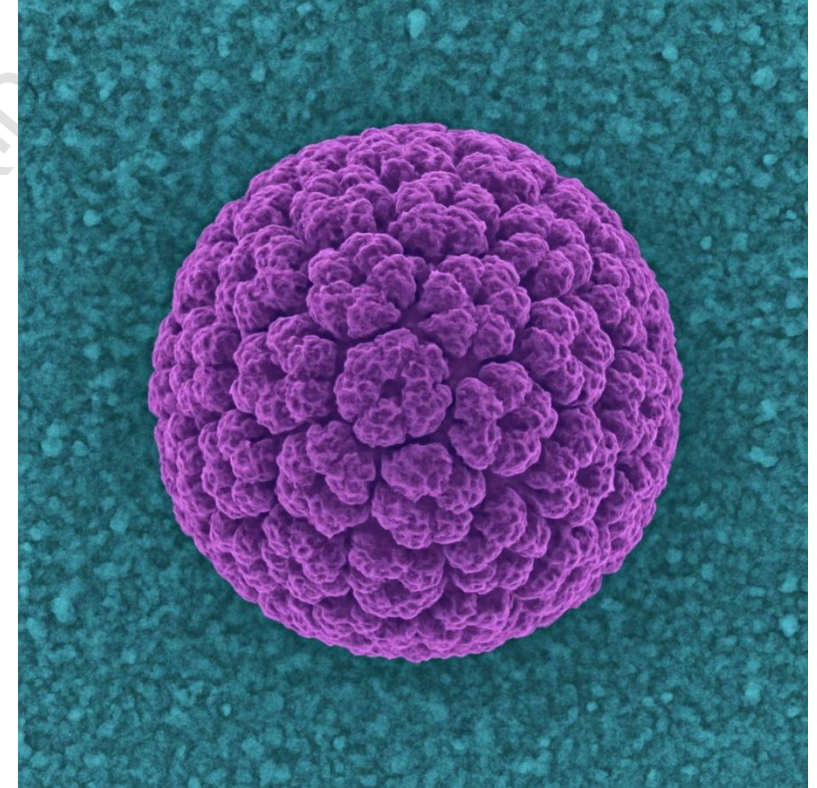
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OVERVIEW

- > 100 different strains of HPV exist (types 16, 18 and a dozen others are oncogenic).
- HPV is generally asymptomatic but can be large and multifocal, producing variable degrees of discomfort and social stigma
- Seroprevalence is incredibly high (nearly 100% of adults)
- Multi-serotype infection/reinfection is common in young, sexually active populations



CLINICAL PRESENTATION

- Mostly asymptomatic
- Warts: multiple, raised, asymmetrical, polymorphic, visible on the anogenital area
- May fluctuate in size and number or regress
- Most warts will regress spontaneously within 1-2 years
- Occasionally bleeding, pruritus, local discharge
- Females: affects cervix, vagina, vulva, anus
- Males: affects anus, penis

GOALS OF THERAPY

- ✓ Symptomatic relief
- ✓ Remove symptomatic lesions
- ✓ Decrease spread to sexual partners
- ✓ Decrease vertical transmission from mother to newborn
- ✓ Reduce transmission of HIV
- ✓ Decrease probability of complications such as infertility, chronic pain, sepsis
- ✓ Prevent disease in uninfected patients

PHARMACOLOGICAL MEASURES

Patient administered: Sinecatchins 10%	• Applied three times daily (up to 250 mg or 0.5 cm of ointment maximum per single dose to all warts) • No need to wash off • Can be used for up to 16 weeks
Patient administered: Imiquimod 3.75% or 5% cream	• 3.75% apply QHS for up to 8 weeks. Less local irritation than 5% cream. • 5% apply QHS 3 times/week divided for up to 16 weeks. • Left on for 8 hours (6-10 hours for 5% cream) and washed off with soap and water • May be up to 8 weeks before warts begin to clear.
MD administered: cryotherapy (liquid nitrogen)	• Adverse effects: pain and skin blistering • If refractory to topical treatment, consider surgery

PREVENTION- VACCINES

Gardasil-9®/Cervarix®

- Polyvalent HPV vaccines
 - Gardasil® -9: type 6, 11, 16, 18, 31, 33, 45, 52, 58
 - Cervarix®: type 16,18
- High efficacy (~96%), safe, should be given to all individuals prior to risk of HPV infection
- Recommended for all persons 9 to 26 years, but older individuals may still get it

Immune function	Age	Immunization Schedule
Healthy	9-20 years of age	1 or 2 doses
Healthy	≥ 21 years of age	2 doses
Immunocompromised/HIV		3 doses

CHANCROID

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OVERVIEW

- Caused by Gram-negative diplococci, *Haemophilus ducreyi*
- Associated with a 10-50-fold increased risk of HIV acquisition
- Often mistaken for syphilis or other GUD
- A chancroid is an initial single papule which may become multiple papules, then a pustule
- It transforms into an erosive painful ulcer that can bleed and can involve the lymph nodes in the groin
- Typically, a chancroid self-resolves

PHARMACOLOGICAL MEASURES

Preferred therapy	• Azithromycin PO x 1 dose • Ceftriaxone IM/IV x 1 dose
Alternative therapy	• Ciprofloxacin PO x 3 days

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THERAPEUTIC TIPS

- “Test of Cure” not required unless symptoms persist
- Important co-factor for HIV transmission and risk of other STIs
- Sexual partners require assessment and treatment if the encounter occurred 10 days from symptom onset
- Abstain from sexual intercourse until the ulcer has resolved
- Use latex or urethane condoms for prevention

PELVIC INFLAMMATORY DISEASE (PID)

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OVERVIEW

- PID commonly refers to a variety of inflammatory disorders of the upper female reproductive tract
- PID also represents a polymicrobial infection that occurs acutely when either vaginal or cervical microorganisms traverse the sterile endometrium and ascend to the fallopian tubes
- *Polymicrobial: N. gonorrhea, C. trachomatis, Mycoplasma hominis, Ureaplasma urealyticum, Bacteroides spp., Prevotella spp, E.coli, Gardnerella vaginalis, H. influenzae*
- Risk factors include: douching, bacterial vaginitis, sex while menstruating, smoking, using intrauterine devices (it is unclear if IUD increases the risk of PID, but it may be prudent to avoid placement when the patient has chlamydial or gonococcal cervicitis)

CLINICAL PRESENTATION

- Abdominal pain, tenderness (after menstrual period)
- Vaginal discharge, menorrhagia, dysuria
- Uterine and adnexal tenderness
- Mucopurulent endocervical exudate
- Chlamydial PID:
 - Slower onset and symptoms are milder than gonococcal disease
 - Chlamydial cases may be more asymptomatic than gonococcal cases

PHARMACOLOGICAL MEASURES

- Incidence of PID can be reduced through active screening of Chlamydia in younger sexually active women
- Treatment depends on severity of the disease
- Data suggests efficacy and long-term complications rates are no different between parenteral and oral therapy
- Inpatients treated empirically after a cervical discharge has been obtained for Gram stain/culture

Inpatient: regimen A	• Cefoxitin IV plus doxycycline IV or PO Continue doxycycline PO after discharge for a total of 14 days of therapy Doxycycline may have pain on infusion and oral doses should be started as soon as possible (IV is Special Access) Optimal coverage of <i>N. gonorrhoea</i> (including resistant strains) and <i>C. trachomatis</i>, possible sub-optimal coverage
Inpatient: regimen B	• Clindamycin IV + gentamicin IV Parenteral therapy may be discontinued after 24 hours if improvement and continue clindamycin PO or doxycycline PO to complete 14 days of therapy Patients with decreased renal function may need dose adjustment of gentamicin Optimal coverage of anaerobes and G-ve enteric rods

PHARMACOLOGICAL MEASURES

Outpatient: regimen A	• Ceftriaxone IM in a single dose + doxycycline PO x 14 days with or without metronidazole PO x 14 days Experts recommend the addition of metronidazole PO x 14 days for anaerobic coverage and treatment of bacterial vaginosis Good to excellent coverage of <i>N. gonorrhea</i> and optimal coverage of <i>C. trachomatis</i>
Outpatient: regimen B	• Levofloxacin PO with or without metronidazole PO for a total of 14 days Adequate but not optimal coverage of <i>N. Gonorrhea</i> and <i>C. trachomatis</i> Lack of anaerobic coverage necessitates use of Metronidazole

MONITORING PARAMETERS

- Substantial clinical improvement is usually seen in 2-3 days and outpatients should be re-evaluated in 2-3 days
- “Test of cure” only required if confirmed chlamydia or gonorrhea
- Follow up for other conditions (pregnancy) or if symptoms return/persist

THERAPEUTIC TIPS

- Sexual partners require assessment and treatment if chlamydia and gonorrhea as per subsections
- **Abstain from sexual intercourse for 7 days after completing**
- Use latex or urethane condoms for prevention of STI causes
- Avoid monotherapy (especially with tetracycline) due to pathogen mix. Tetracyclines have suboptimal activity against G^{-ve} aerobic and anaerobic organisms and against some strains of *N. gonorrhoea*

SEXUALLY TRANSMITTED ECTOPARASITES (PUBIC LICE AND SCABIES)

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OVERVIEW

- Pubic lice (*Phthirus pubis*) and scabies (*Sarcoptes scabiei*) transmitted by close/intimate contact and fomites
- Pubic lice, like head lice, lay eggs (nits) on hair shafts. Nymphs and adult lice feed on host ("blood meal") to survive. Adult lice are ~1.5 mm long
- Scabies burrow under the epidermis and lay eggs in the burrow. Adult scabies are very small (<0.5mm)



CLINICAL PRESENTATION

Men and women

- Scabies commonly infect the hands (between fingers) and toes (between toes), axilla, belly button and groin. Incubation time: 3 weeks (more rapid in reinfection)
- Pubic lice typically infest pubic hair but can also attach to armpit hair, beards, eyelashes/brows
- Scabies and lice cause intense itching in the area of infection
- Itching can lead to excoriation and secondary infection

Immunocompromised patients

- May present with scabies “hyperinfection” known as Norwegian scabies/crusted scabies

NON-PHARMACOLOGICAL MEASURES

Pubic lice

- Wash affected clothing/bedding in hot water, dry clean or seal in plastic bag for 14 days
- Vacuum mattresses
- Comb nits from hair shafts using a lice comb (in conjunction with a pediculicide)

Scabies

- Wash affected clothing/bedding in hot water, dry clean, or seal in plastic bag for 3-7 days
- Vacuum mattresses

PHARMACOLOGICAL MEASURES- PUBIC LICE

Pubic lice: preferred treatment	• Permethrin 1% cream rinse: apply to wet hair, leave on for 10 minutes, then rinse. Repeat in 7-10 days if live lice seen • Mixed pyrethrins (0.33% pyrethrin-piperonyl butoxide) shampoo: apply to dry hair and let sit for 10 minutes, then rinse. Repeat in 7-10 days Avoid mixed pyrethrins if allergic to chrysanthemum (note controversy re: permethrin) Avoid eye contact Avoid all products if eyelashes/eyelids are involved (refer to physician/nurse practitioner)
Pubic lice: alternative treatment	• Lindane no longer commercially available

PHARMACOLOGICAL MEASURES- SCABIES

Scabies: preferred treatment

- Permethrin 5% cream: applied to body (from chin down to toes, including under fingernails and toenails)

Leave on for 8-14 hours then rinse. Reapply in 1-2 weeks

Scabies: alternative treatment

- Sulphur 5-10% in petrolatum applied nightly for 3 nights, leave on for 24 hours, wash off prior to next application
- Ivermectin (salvage regimen for Norwegian scabies) PO x 3-7 doses in addition to topical permethrin

MONITORING PARAMETERS

- “Test of Cure” is NOT recommended after completion of therapy
- Itching may persist for days to weeks after effective therapy
- Risk for secondary infections, have patients report pain, swelling, fevers, etc.
- Norwegian scabies requires close follow up and is often associated with systemic secondary bacterial infections

THERAPEUTIC TIPS

- Sexual partners/intimate contacts of patients with both pubic lice and scabies (even if asymptomatic) within 30 days require treatment
- All members of the same household require treatment for scabies

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CHANGE LOG

April 2025

- Slide 11: Updated gonorrhea treatment guidelines in line with new Canadian guidance
- Slide 18: Removed PO erythromycin as discontinued
- Slide 26: Updated ceftriaxone from 14 day duration to 10-14 days as per CDC, rxtx. Placed note in speaker notes that probenecid is d/ced, may be accessed as SAP drug
- Slide 33: Added '6-10 hours for 5% cream'
- Slide 37: Removed PO erythromycin as discontinued
- Slide 43: Updated outpatient regimen A based on Canadian guidelines

May 2025

- Slide 13: Updated when to perform culture and NAAT based on Gov of Canada guidelines
- Slide 49: Changed recommendation for storing items in sealed bags from 7 days to 14 days as per UpToDate
- Slide 51: Clarified application instructions for sulfur
- Slide 53: Clarified information for pubic lice and scabies regarding close contacts (info placed in brackets)

June 2025

- Added cases throughout
- Slide 6: clarified risk factors/risk groups
- Slide 11: clarified management of gonorrhea in setting of unknown chlamydia

October 2025

- No content changes made

November 2025

- Slide 21: Updated timeframe for repeat screening to 3 months
- Slide 59: Updated references

February 2026

- Slide 58: Removed "if live lice is visualized" for pubic lice, as sexual partners need to be treated regardless to prevent transmission
- Slide 59: Updated references